Baskin Engineering

FIREFLY

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PROBLEM

Remote and high fire-risk areas lack the sensor infrastructure necessary to monitor environmental conditions during the critical pre-fire window.

SOLUTION

Drone nodes detect fire/smoke, send GPS telemetry, and help estimate fire location on a live ground-station map.

OBJECTIVES

- Detect fire/smoke from aerial video
- Transmit drone GPS, heading, and alert status
- Locate fire using 2+ drone observations
- Display alerts on a live map

TARGET USERS

- Wildfire operations coordinators
- Fire-response and forestry field crews
- Environmental researchers
- Future engineering teams

</> SOFTWARE

Ground Station Interface

- Displays drone positions, alerts, and estimated fire markers

Vision Pipeline

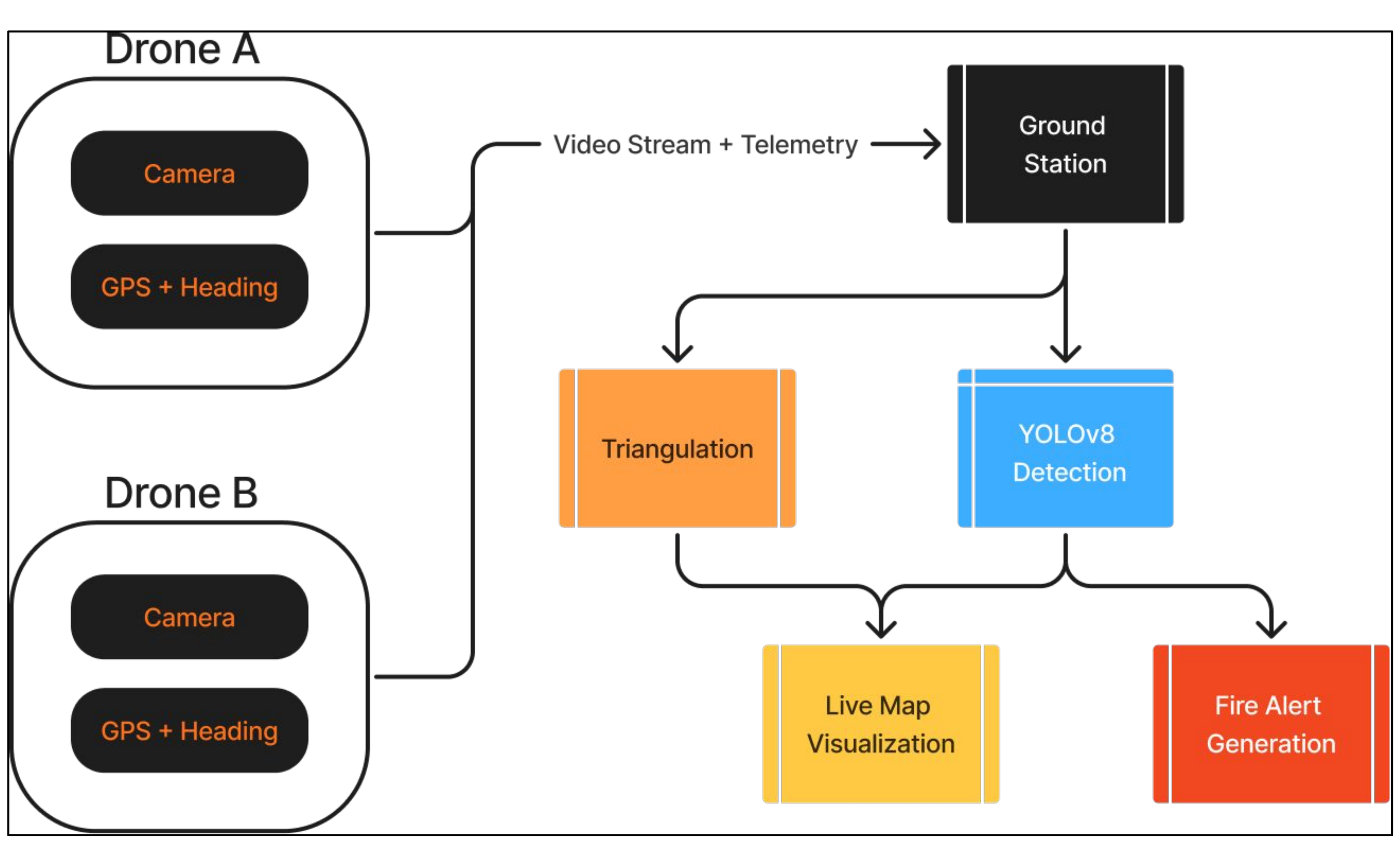
- YOLOv8 fire and smoke detection

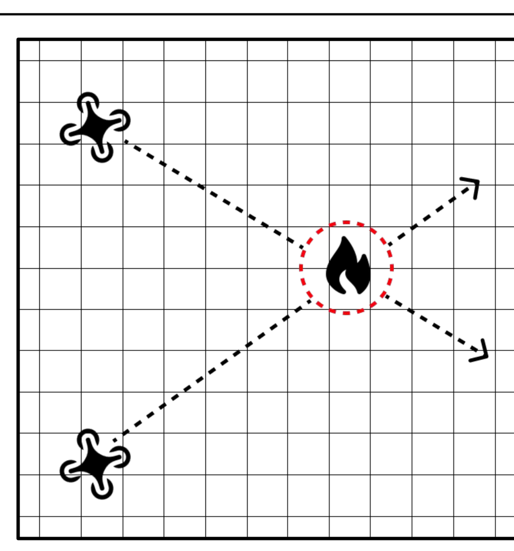
Localization

- Estimates GPS coordinates of detected fire events

Telemetry Backend

- Parses real-time telemetry messages
- Updates active drone and fire-event state





HARDWARE

Drone Node

- Flight/telemetry microcontroller
- Camera video module
- GPS receiver for position data

Wireless Communication

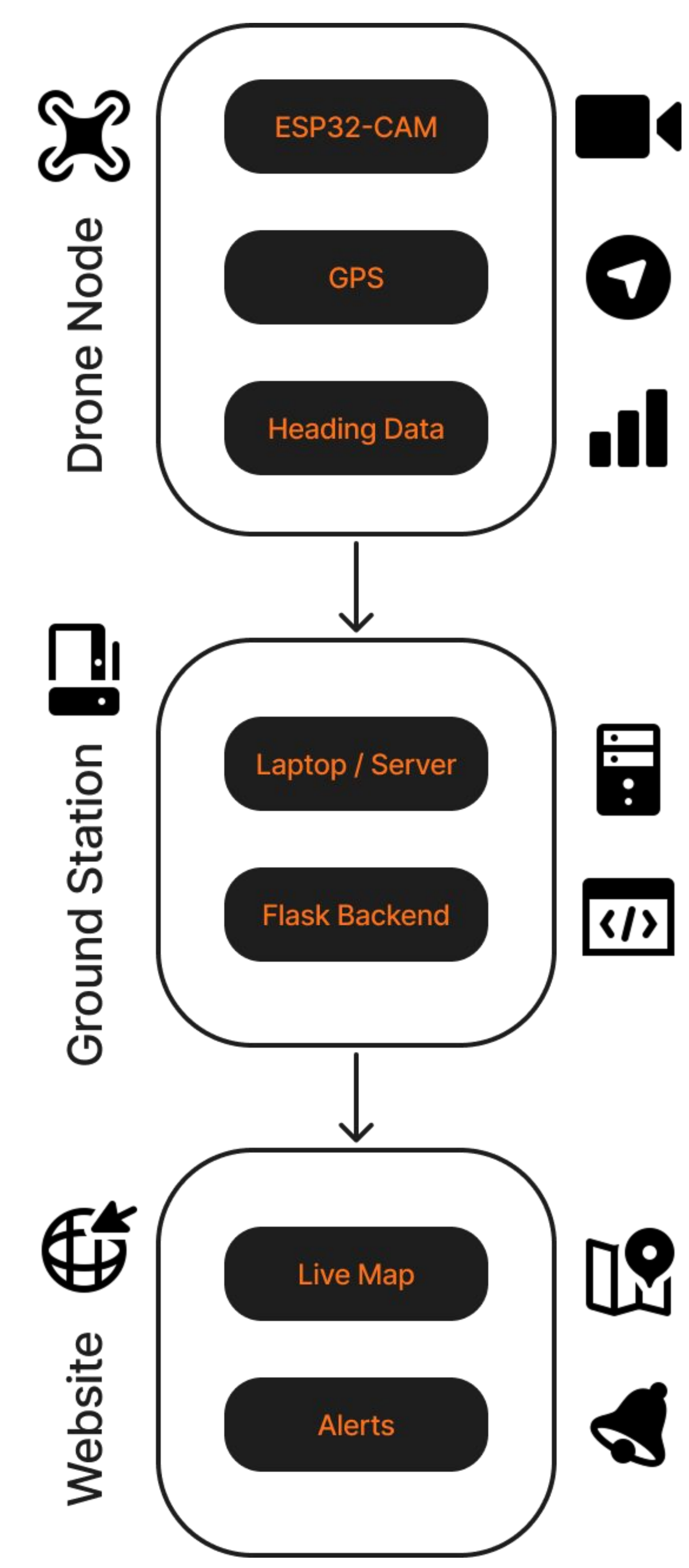
- broadcast telemetry between drone nodes
- Wi-Fi connection to ground station

Ground Station

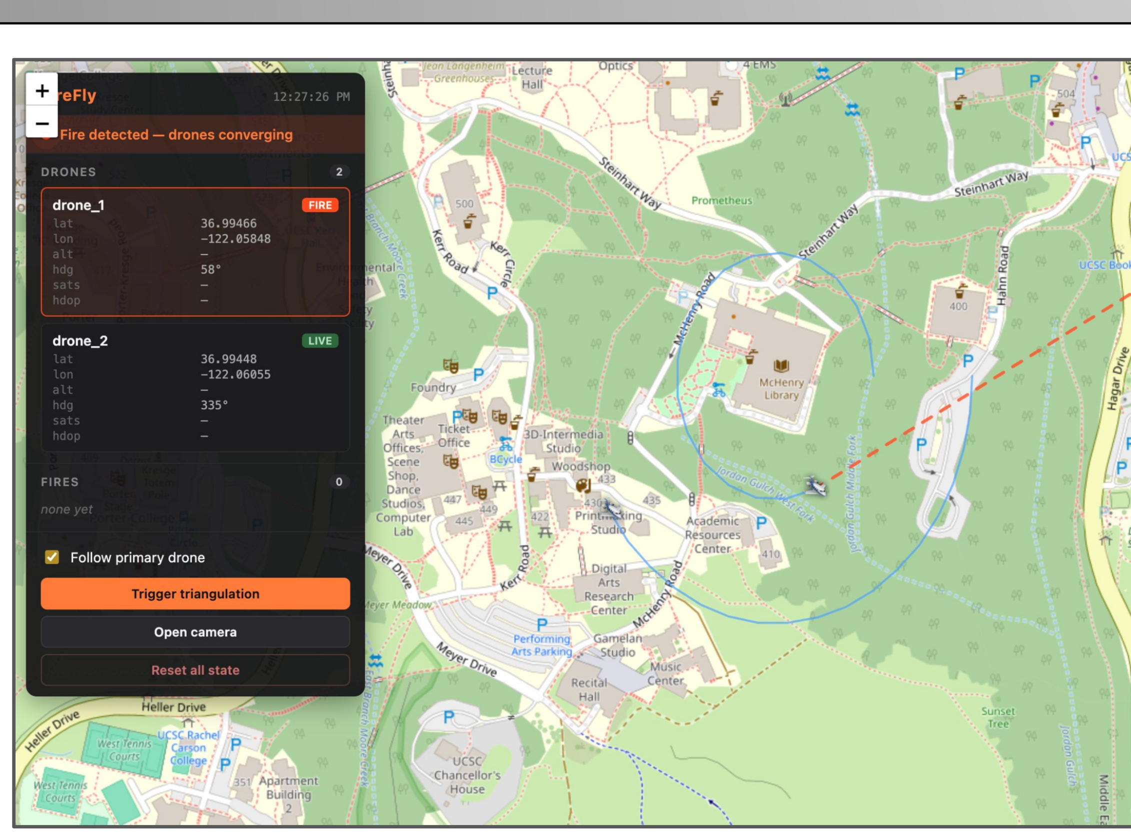
- Receives telemetry and fire alerts
- Displays live map and detection results

Sensor Data Collected

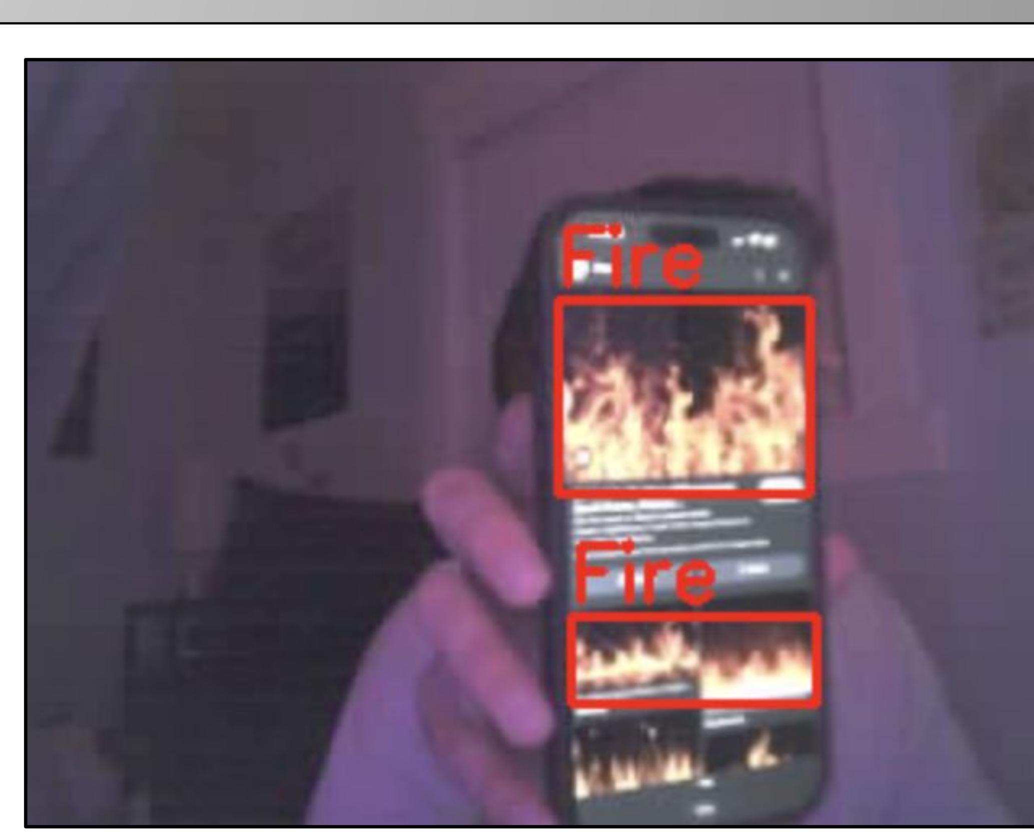
- Camera stream for fire/smoke detection
- GPS coordinates



DEMO AND VALIDATION



Ground Station Map



Fire Detection View

FIREFLY WEBSITE

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